

Product Requirements Document (PRD)

FlowLens – Warehouse Workflow Intelligence Platform

Version: 1.0

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Product Type: SaaS Web Application

Target Users: Operations Managers, Warehouse Supervisors, Quality Assurance Teams, Regional Operations Heads

1. Overview

Problem Statement

A grocery delivery startup currently tracks operational metrics such as order preparation time, packing accuracy, and delivery complaints. However, these metrics exist in silos, making it difficult for operations teams to identify which warehouse workflows are responsible for downstream failures. As a result, issues are addressed only after customers are affected, leading to increased complaints, higher operational costs, and reduced customer satisfaction.

Proposed Solution

FlowLens is an Operational Intelligence Platform that provides end-to-end visibility into warehouse operations. It tracks every order through each processing stage, identifies workflow bottlenecks, predicts delivery failures before dispatch, and recommends corrective actions using real-time analytics and AI.

2. Goals

Business Goals

- Reduce delivery complaints by identifying root causes early.
- Improve warehouse operational efficiency.
- Increase packing accuracy.
- Reduce order processing delays.
- Enable data-driven operational decisions.

Product Goals

- Track every order throughout the warehouse workflow.
- Detect bottlenecks automatically.
- Predict high-risk orders before dispatch.

- Recommend corrective actions in real time.
 - Provide operational transparency across warehouses.
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3. Target Users

Operations Manager

Responsibilities:

- Monitor warehouse performance.
- Reduce complaint rates.
- Improve workflow efficiency.
- Allocate workforce effectively.

Pain Points:

- Cannot identify root causes of failures.
 - Manual investigation takes significant time.
 - Limited visibility across multiple warehouses.
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Warehouse Supervisor

Responsibilities:

- Manage daily operations.
- Track employee productivity.
- Resolve bottlenecks quickly.

Pain Points:

- Delays are identified too late.
 - Difficult to monitor every process simultaneously.
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Quality Assurance Team

Responsibilities:

- Ensure packing quality.
- Reduce damaged or incorrect deliveries.

Pain Points:

- Cannot identify recurring quality issues.
 - No historical workflow analysis.
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4. User Stories

Operations Manager

- As an Operations Manager, I want to monitor every warehouse in real time so that I can quickly identify operational issues.
 - As an Operations Manager, I want to know which workflow stage causes the most customer complaints so that I can prioritize improvements.
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Warehouse Supervisor

- As a Supervisor, I want alerts when order preparation exceeds SLA so that I can intervene immediately.
 - As a Supervisor, I want to view employee workloads to balance assignments.
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QA Team

- As a QA Analyst, I want to identify packing-related issues before dispatch.
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5. Functional Requirements

FR-1 User Authentication

Users shall:

- Login securely.
- Reset passwords.
- Access role-based dashboards.

Roles:

- Admin
 - Operations Manager
 - Warehouse Supervisor
 - QA Team
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FR-2 Warehouse Management

System shall:

- Register warehouses.
- Define warehouse zones.

- Track active warehouses.
 - Configure operational capacity.
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FR-3 Order Lifecycle Tracking

Each order shall include:

- Order ID
- Customer ID
- Warehouse
- Assigned Employee
- Current Stage
- Stage Timestamp
- Processing Time
- SLA Status

Workflow:

Order Received

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Picking

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Packing

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Quality Check

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Dispatch

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Delivery

FR-4 Workflow Monitoring

System shall monitor:

- Time spent in each stage
- Employee handling stage
- Queue length

- Waiting time
 - Processing delays
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FR-5 Bottleneck Detection

System shall automatically detect:

- Long queues
 - High waiting time
 - Slow processing stages
 - Frequently failing workflow steps
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FR-6 Complaint Management

Store complaints including:

- Complaint Type
- Complaint Time
- Order ID
- Warehouse
- Delivery Executive
- Root Cause

Complaint Types:

- Wrong Item
 - Missing Item
 - Damaged Item
 - Late Delivery
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FR-7 Root Cause Analysis Engine

System shall:

- Trace complaints back through workflow history.
 - Identify warehouse responsible.
 - Identify workflow stage responsible.
 - Identify responsible shift.
 - Identify employee involved.
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FR-8 AI Risk Prediction

Before dispatch:

System calculates:

- Risk Score (0–100)

Inputs:

- Preparation Time
- Packing Accuracy
- Queue Size
- Employee Performance
- Warehouse Load
- Historical Complaint Rate

Outputs:

- Low Risk
 - Medium Risk
 - High Risk
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FR-9 Intelligent Recommendations

System recommends actions such as:

- Add additional packing staff.
 - Reassign workers.
 - Trigger quality inspection.
 - Delay dispatch for verification.
 - Open additional packing station.
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FR-10 Alerts & Notifications

Generate alerts for:

- SLA Breach
- Packing Accuracy Drop
- High Complaint Rate
- High Warehouse Load
- High Risk Orders

Notifications via:

- Web Dashboard
 - Email
 - SMS (optional)
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FR-11 Reporting

Generate reports for:

- Warehouse Performance
 - Employee Productivity
 - Complaint Trends
 - SLA Compliance
 - Workflow Efficiency
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6. Non-Functional Requirements

Performance

- Dashboard updates within 2 seconds.
 - Support 100,000+ orders/day.
 - API response time below 300 ms.
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Availability

- 99.9% uptime.
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Security

- JWT Authentication
 - Role-Based Access Control (RBAC)
 - HTTPS encryption
 - Audit logs
 - Data encryption at rest
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Scalability

- Multi-warehouse support.
 - Horizontal backend scaling.
 - Cloud-native deployment.
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7. System Architecture

Frontend

- React.js
- TypeScript
- Material UI

- React Flow
- Apache ECharts

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Backend

- Node.js
- Express.js
- REST APIs
- WebSockets

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AI Engine

- Python
- FastAPI
- Scikit-learn / XGBoost

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Database

- PostgreSQL
- Redis
- MongoDB

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Cloud

- Docker
- Kubernetes
- AWS

8. Database Entities

Users

- User ID
 - Name
 - Role
 - Email
 - Password
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Warehouses

- Warehouse ID
 - Name
 - Location
 - Capacity
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Orders

- Order ID
 - Customer ID
 - Warehouse ID
 - Current Status
 - Risk Score
 - Processing Time
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Workflow Logs

- Order ID
 - Stage
 - Employee
 - Start Time
 - End Time
 - Duration
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Complaints

- Complaint ID
 - Order ID
 - Complaint Type
 - Root Cause
 - Resolution Status
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9. Success Metrics (KPIs)

- Delivery Complaint Rate
- Average Order Preparation Time
- Packing Accuracy (%)
- On-Time Dispatch Rate
- Workflow Bottleneck Frequency
- AI Prediction Accuracy
- SLA Compliance (%)
- Average Complaint Resolution Time
- Warehouse Utilization
- Employee Productivity

10. Future Enhancements

- Computer Vision for packing verification.
- IoT integration with warehouse sensors.
- Demand forecasting using ML.
- Workforce scheduling optimization.
- Voice-enabled warehouse assistant.
- Mobile application for supervisors.
- Predictive maintenance for warehouse equipment.
- Digital Twin simulation for warehouse operations.

11. Assumptions & Constraints

Assumptions

- Warehouses digitally record every workflow stage.
- Complaint data is available for each order.
- Employees use barcode/RFID scanning during processing.
- Internet connectivity is available within warehouses.

Constraints

- Initial release supports only grocery warehouse workflows.
- AI predictions depend on historical operational data quality.
- Real-time updates require stable network infrastructure.

12. Expected Business Impact

FlowLens transforms warehouse operations from reactive monitoring to proactive decision-making. By connecting operational workflows with customer outcomes, the platform enables managers to identify bottlenecks, predict failures before dispatch, and implement corrective actions in real time. This leads to reduced complaint rates, improved operational efficiency, higher packing accuracy, better SLA compliance, and an overall improvement in customer satisfaction while lowering operational costs.